Status	Finished
Started	Saturday, 10 October 2026, 3:05 PM
Completed	Saturday, 10 October 2026, 3:13 PM
Duration	7 mins 43 secs
Grade	**13.00** out of 15.00 (**86.67**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The search continues from the beginning
b. The list is sorted
c. The search stops
d. The search continues in the left or right sublist ✓

Question **2**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It works on unsorted lists
b. It always starts from the beginning of the list
c. It can be applied only if the list is sorted ✓
d. It compares elements sequentially

Question **3**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the search algorithm finishes
b. When the list is sorted
c. When the list contains duplicate elements
d. When the element is found in the list ✓

Question **4**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. In the left sublist
b. In the middle of the list
c. In the right sublist ✓
d. At the beginning of the list

Question **5**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(1)
b. O🥐
c. O(log n) ✓
d. O(n log n)

Question **6**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By dividing the list length by two ✓
b. By starting from the first element
c. By using a hash function
d. By comparing each element sequentially

Question **7**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

a. Sort the list
b. Compare the target element with the middle element in the list ✓
c. Divide the list into two equal parts
d. Compare the target element with the first element in the list

Question **8**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It does not work with negative numbers
b. It is slower than linear search for small lists
c. It can only be applied to large lists
d. It requires the list to be sorted ✓

Question **9**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list is unsorted
b. When the list contains duplicate elements
c. When the list is very small
d. When the list is sorted ✓

Question **10**

Incorrect

Mark 0.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The search ends successfully
b. The list is sorted
c. The search continues in the right sublist
d. The search continues in the left sublist ✗

Question **11**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

a. Dynamic search
b. Binary search
c. Hashing
d. Linear search ✓

Question **12**

Incorrect

Mark 0.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 89 and 94

b. 90 and 100

c. 94 and 99 ✗

d. 90 and 99

Question **13**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. When the item is not the array at all

b. Item is the last element in the array or item is not there at all

c. When the item is somewhere in the middle of the array ✓

d. When the item is the last element in the array

Question **14**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O(n log n)

b. O(log n)

c. O(1) ✓

d. O👎

Question **15**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

a. Jump search

b. Hashing

c. Binary search

d. Linear search ✓